3.1 Give a single intensity transformation function for spreading the intensities of an image so the lowest intensity is 0 and the highest is $L-1$.

3.5 In general:

(a) * What effect would setting to zero the lower- order bit planes have on the histogram of an image?

(b) What would be the effect on the histogram if we set to zero the higher-order bit planes instead?

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Let the input gray level be r with minimum r_{min} and maximum r_{max} in the image, and the desired output range be $[0, L-1]$.

$$S = (L-1)\frac{r - r_{min}}{r_{max} - r_{min}}$$

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(a) Remove fine, subtle variations (often noise or small details). The histogram collapses onto gray levels that are multiples of 2^k (where K LSBs are zero-ed) so it becomes spiker with fewer distinct bins; overall shape is similar but with mild shift darker (because removed LSBs contributed a small positive average).

(b) Discards most of the significant visual information (brightness/contrast). The histogram is squeezed into a narrow range near black (only the low bits remain) yielding a very low-contrast, dark image, again with peaks separated by powers of two tied to the remaining bit depth.